

Detailed Forecast & Analysis Report

Predicted Load: 116.09301813532554 MWh

Uncertainty: ± 3.855839969120463 MWh

AI Executive Summary & Reasoning

The energy consumption forecast for the upcoming period is projected at **116.09 units**, with an associated uncertainty of **± 3.86 units**. This prediction is derived from a robust ensemble model, combining outputs from XGBoost, LightGBM, and CatBoost algorithms. CatBoost significantly influenced the final forecast with a 65% weighting, contributing a prediction of 113.14 units. LightGBM, weighted at 30%, suggested a higher value of 122.44 units, while XGBoost, with a 5% weight, predicted 116.40 units. The ensemble approach leverages the strengths of these diverse models, enhancing the reliability and stability of the overall forecast by balancing individual model biases.

The primary driver for this forecast, particularly its downward adjustment relative to recent historical consumption, is the **time-of-day factor**, which exerted a substantial negative contribution of -9.37 units. This suggests that the specific target hour (14:00) and its associated consumption patterns, potentially influenced by the model's interpretation of a 'weekend-like' consumption profile (`is_weekend: 1`) despite the calendar day being a Friday, are leading to an expected dip in demand. Historical consumption patterns also play a role, with energy demand from 24 hours prior (128 units) and 12 hours prior (125.4 units) providing a moderate upward influence. However, this was not enough to offset the dominant time-of-day effect. The absence of a significant contribution from `weather_temp` indicates that temperature is not a primary factor influencing this specific forecast. Overall, the forecast reflects an anticipated decrease in demand compared to the recent 24-hour rolling mean of 130.6 units, primarily due to the expected lower consumption typical for this specific time and day-type.

Key Factor Breakdown

- hour: 14
- day_of_week: 5
- month: 6
- is_weekend: 1
- lag_24h: 128

- lag_12h: 125.4
- roll_mean_24h: 130.6
- roll_std_24h: 4.2